



Jury priors and observable defendant characteristics

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ABSTRACT

Although prohibited, jurors sometimes condition, consciously or subconsciously, their belief that a defendant is guilty on the defendant's race or ethnicity or other observable characteristics. This can be viewed as a juror forming a prior or pre-trial/evidence disclosure belief of guilt. In doing this, they rely on their perceptions of education, socio-economic status, religion, beliefs, networks, etc. for the defendant's race (or other observable characteristic) and how they perceive those to influence the probability the defendant is guilty. This is consistent with aversive discrimination, which suggests that people want to be egalitarian and not condition on race but have a tendency to base decisions on factors that are discriminatory when race is not salient. When this prior or pre-trial/evidence disclosure belief of guilt overestimates the prior probability of guilt for those in the minority group, it underestimates the prior probability of guilt for those in the majority group. Prohibiting conditioning on observable defendant characteristics can be viewed as requiring the use of the population prior/pre-trial probability of guilt. Conditions for when such prohibition improves accuracy are provided. While it is difficult to effectively prohibit this, studies of aversive discrimination suggest that making race salient in a trial can reduce implicit bias on race. So these results may provide some guidance on when such activity should be permitted.

1. Introduction

Although in the U.S. jurors are not to condition on a defendant's race or ethnicity, anecdotal evidence suggests that jurors do so when it is observable. Considerable forensic psychology work documents this. See, for example, [Bodenhausen \(1988\)](#). A juror conditioning on the defendant's race or ethnicity is consistent with a view that criminal propensity varies by race or ethnicity. How race or ethnicity influences criminal propensity is not observable; jurors may infer this, consciously or subconsciously, based on other factors they may know or believe they know.² Clearly, guilt does not stem directly from race or ethnicity.

It's relatively common for activists to emphasize educational attainment, career successes, and positive attributes of minority groups that may not fit with stereotypes. This can encourage young minorities and provides information about the minority population. Such discrimination can have efficiency effects on the primary activity of those who may be discriminated against. While these issues and fairness

considerations provide compelling arguments against allowing jurors to condition on race or ethnicity, some argue a prohibition on it results in information being disregarded.

Bayesian updating based on evidence presented at trial is a natural way to model a jury's decision making. However, each juror has an individual prior belief the defendant is guilty that is influenced by the juror's experiences and background. The extent of authentic experience with members of a particular minority group likely varies. Some recent papers, such as [Lundberg and Mungan \(2022\)](#) and [Sullivan \(2019\)](#), have suggested the permissibility of conditioning on observed defendant characteristics may be desirable, at least from an information standpoint for the case at hand.³ Naturally, these rely on jurors having an accurate sense of the true/correct prior or pre-trial/evidence disclosure probability the defendant is guilty for each group, which empirically is not observed since true guilt or innocence are not observed. The focus in this paper is on how jurors might form,

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² There is a large body of psychology literature on subconscious discrimination and an economics literature as well. See for example, [Bertrand et al. \(2005\)](#).

³ Lundberg and Mungan show that, when the jury beliefs are rational, any prohibition on the use of defendant characteristics as a form of evidence will never simultaneously reduce conviction rates for the innocent and increase conviction rates for the guilty. They note, "Specifically, the jury (plural) assign a prior belief of guilt to defendants based on some observable or inferable characteristic, such as race, criminal history, or occupational status. We note the model also applies to victims or witnesses who testify at trial—if the jury assign them to groups of different credibility, then, by extension, the jury assign defendants to groups of different criminal likelihood". They also show that juror's bias may be a reason to not allow jurors to use defendant characteristics as a form of evidence.

consciously or subconsciously, a belief that the defendant is guilty prior to the disclosure of case-specific evidence at trial. In Bayesian models of evidence disclosure, such a prior belief is usually taken as given.

A potential problem with jurors conditioning their priors on an observed characteristic is that a defendant belongs to many different groups. For example, in addition to a particular racial or ethnic identity, the defendant has a specific socio-economic background, an education level, different networks, religious affiliation or beliefs, and so on.⁴ It's possible that a juror may use the defendant's membership in one group, such as race, to infer how likely it is that the defendant belongs to other groups, which the juror doesn't observe.

Such inference can be incorrect or biased and lead to the use of an incorrect prior. [Lundberg and Mungan \(2022\)](#) show when the jury sufficiently over-weighs differences across group characteristics, an impermissibility rule improves accuracy in verdicts for both guilty and innocent defendants. The use of an incorrect prior can distort incentives in primary activity as allowing character evidence does in [Sanchirico \(2001\)](#), and [Schrag and Scotchmer \(1994\)](#).

In *Peña-Rodriguez v. Colorado*, 137 U.S. 855 [2017], a juror referred to as "HC" made racist comments about the defendant to other jurors. The statements HC made indicated that HC viewed the defendant as being similar to those of the same ethnicity that HC had encountered while working in law enforcement. HC likely made assumptions, based on the defendant's ethnicity, about his socio-economic background, education level, and moral standards. In that case, it seemed to be a conscious inference, but in many instances it may be subconscious. There is a large literature dealing with implicit bias in trials and the judicial system more generally.

For some time, the legal psychology literature has demonstrated that intentional discrimination is generally not the cause of discrimination nowadays. Instead, discrimination is often the result of subconscious biases. See for, for example, [Wang \(2004\)](#), [Armour \(1995\)](#), [Krieger \(1995, 1998\)](#), and [Greenwald and Krieger \(2006\)](#). [Greenwald and Krieger \(2006\)](#) note, regarding Implicit Association Test (IAT) data, "The results shown in Table 2 strongly suggest that any non-African American sub-group of the United States population will reveal high proportions of persons showing statistically noticeable implicit race bias in favor of EA [European Americans] relative to AA [African Americans]". It's noted that many people wish to be egalitarian and not discriminate, but there are situations where subconscious biases appear.⁵ Psychology studies indicate that when race is salient, instead of being obscured, racial bias by Whites is reduced. [Sommers and Ellsworth \(2000, 2001\)](#) are seminal work in this direction. This notion is often referred to as aversive discrimination. Often the bias is the result of relying on an attribute that is correlated with race. Further, it has been suggested that attempts at a "color-blind" approach can lead to increases in this type of discrimination.⁶ [Armour \(1995\)](#) suggests courts allow for "rationality-enhancing" group references, which would make race more salient and thus reduce the scope for implicit bias. However, based on examples discussed by [Armour](#), it seems that is not always an available option as judges are reluctant to permit it.

There is a sense in which, especially when race is not salient, that a decision maker may base a decision on a factor that is viewed as being legitimate and nondiscriminatory, but it has the effect of discriminating.⁷ [Armour \(1995\)](#) notes, "In contrast, when the situation is normatively ambiguous, or when a nonrace-related justification is

handy, the covert antiblack attitudes and beliefs of aversive racists find expression in racial discrimination".

Stereotypes, which may or may not be accurate, can play a role.⁸ There is much evidence that people tend to categorize others' behavior to help make sense of it. In particular, it is common for people to make inferences about someone's education or socio-economic status.⁹ [O'Flaherty and Sethi \(2019\)](#) describe how in 2016 Tamika Cross, a young, Black, female physician, was initially prevented from assisting an unresponsive passenger on a flight. Cross was told that "the flight crew was looking for 'actual physicians or nurses or some type of medical personnel.'"¹⁰ Following concern on social media about the treatment of Cross, the airline said that requesting medical identification was consistent with training flight attendants receive. It seems such inferences are not isolated occurrences.

[Deloria's 2004](#) book *Indians in Unexpected Places* provides accounts of Native Americans doing unexpected things, with a focus on the late nineteenth century and early twentieth century, such as directing and acting in movies, playing professional baseball, singing opera, and driving automobiles. This compels people to reconsider their current views about Native Americans. This certainly focuses on education and socio-economic status.

Another example is Black people who have been either arrested or shot by police at their own homes. The 2009 arrest of Professor Henry Louis Gates, Jr. is well known. See, for example, [Pilkington \(2009\)](#). After Kazeem Oyeneyin's security system accidentally triggered, he was handcuffed and put in a police car while trying to explain that it was his house. When a police sergeant who knew him arrived, he was released.¹¹ [Vielmetti \(2022\)](#) describes another case in which a police officer "violated a Black man's constitutional rights when he entered a home and arrested the man based on a neighbor's racial suspicion, a federal judge has ruled". Other examples with deadly outcomes include that of Sonya Massey.¹² In many of these situations, it seems race may have been a factor in beliefs being formed about a Black person's education or socio-economic status, and whether they live in their own house.

It's consistent with aversive discrimination for a juror to observe a defendant's race and use that to form a prior/pre-trial belief of the defendant's guilt. There is much anecdotal and psychological evidence that race is used, perhaps implicitly, in assessing another's level of education and socio-economic status. Education and socio-economic

The normative ambiguity studies show that racially biased treatment and legitimate, nondiscriminatory justifications are likely to coexist in many cases. More specifically, they show that the existence of a legitimate justification for a negative decision does not necessarily discredit racial bias as an explanation for that decision. The presence of such a justification may, instead, be cause to suspect that the decision in fact was racially biased, because racial discrimination today seems most likely to occur through the racially biased application of a nondiscriminatory reason.

⁸ [Krieger \(1995\)](#) notes, "Stereotypes are correlational constructs". There is a large body of work suggesting that stereotypes can be self-confirming. See, for example, [Agan \(2024\)](#) and [O'Flaherty and Sethi \(2019\)](#).

⁹ [Agan \(2024\)](#) categorizes discrimination as either preference-based or belief-based. Belief-based discrimination includes accurate statistical discrimination and inaccurate-belief based discrimination. [Agan](#) emphasizes that although accurate statistical discrimination may be rational from the decision maker's perspective, it is typically illegal, such as in the criminal setting studied here.

¹⁰ [Hawkins \(2016\)](#) contains further details.

¹¹ The discussion here is based on [Hutchinson \(2019\)](#). Oyeneyin was armed when he answered the door and informed police that he had a gun before answering the door. He said "Twenty minutes later, I just hear these loud noises... So, I come down my steps, I grab my gun because I don't know who's in the house".

¹² See, for example, [Lenenson et al. \(2024\)](#).

⁴ [Bull and Watson \(2019\)](#) study a setting where these differences might influence evidence disclosure at trial. The simpler idea here is that jurors may internalize these features to determine the prior probability of guilt.

⁵ [Roberts \(2012\)](#) discusses the prevalence of implicit bias even among those who aim to be unbiased, and proposes the use of the IAT for jurors in an educational manner.

⁶ See, for example, [Davidio et al. \(2016\)](#).

⁷ [Wang \(2004\)](#) notes

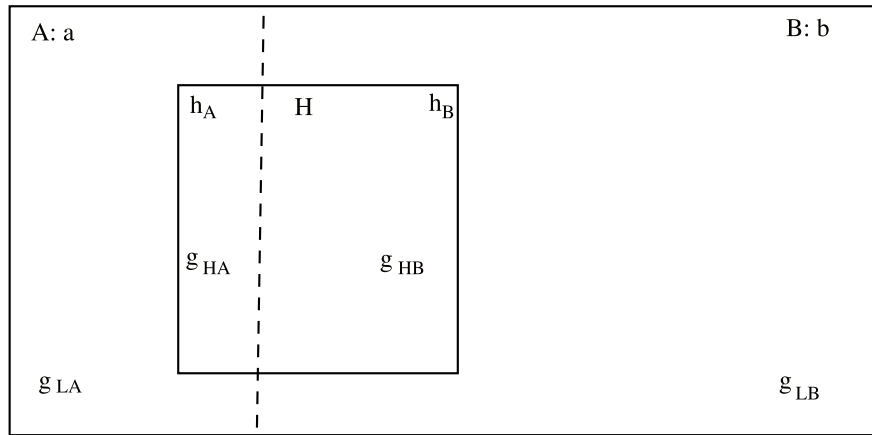


Fig. 1. A graphical representation of groups.

status fit with factors that aversive discrimination would place weight upon.

The exercise here assumes that in addition to belonging to a racial (or other observable) group, the defendant also belongs to some other group, such as determined by education level or socio-economic status, and the rate of belonging to that other group is used by jurors to form a prior probability the defendant is guilty. That is, since the rate of belong to the other group (high education or high socio-economic status) typically varies across race, the juror engages in aversive discrimination.

I assume that the jury uses the correct fraction of each racial group that is in the other/high education group but uses the overall population probability of guilt for the other/high education group.¹³ Naturally, other methods of estimating a prior probability are possible, and some are discussed in the Appendix. However, this approach assumes that the only inaccuracy in the juror's estimate is the use of the population probability of guilt given the education level. I suggest this is appealing for the following reasons. (1) Use of the correct/true probability of guilt given both race and education level would be direct accurate belief-based discrimination which seems at odds with aversive discrimination. Further, use of the population value allows for jurors having a sense of how education influences the probability of guilt regardless of race. (2) While jurors are unlikely to know the true/correct values of any of the probabilities, using the true/correct probability of a member each race being highly educated likely captures the qualitative differences in juror estimates. (3) This is more structured than allowing for mis-estimates of both probabilities, which would make it easier to obtain the kinds of results I'm interested in.

The aim of the paper is to understand whether and when, in the context of a simple model of aversive discrimination where jurors can, perhaps implicitly, condition on differences in education (or socio-economic status) across races, prohibiting conditioning on the defendant's race can improve the accuracy of juror decision making for the case at hand. While it may be difficult in practice to prevent jurors from conditioning on the defendant's race, the large body of work on aversive discrimination suggests that making race salient can have a similar effect. Naturally, judges and attorneys have concerns about allowing the discussion of race. See, for example, Armour (1995). So, the results reported here may provide some guidance on when to do so.

The paper is organized as follows. The next section presents the simple model of different groups. Section 3 presents juror estimation of priors and the analysis. Section 4 contains a brief policy discussion and conclusion. The Appendix contains proofs of results and a brief discussion of other modeling approaches.

2. Simple model

Suppose there are two different sets of groups that an individual can belong to: A or B, and H or L. It may be helpful to think of A as a visible minority group and B as the rest of the population or the majority group; a denotes the number in A and b denotes the number in B. Although we may think of A as a minority and this is helpful for exposition, it need not be that $a < b$ in general. H and L can be viewed as high and low education, respectively; h denotes the number in H and ℓ denotes the number in L.¹⁴ In total there are n individuals in society, and there are g guilty individuals. Subscripts are used to denote the numbers of h and g belonging to each group. For example, g_H denotes the number of guilty individuals in H, and g_{HA} denotes the number of guilty individuals who belong to both H and A. So, for example, $g_L = g_{LA} + g_{LB}$. The language "guilty" to describe g is to convey intuition. The g values and prior capture the criminal propensity of different groups. The distribution of people is represented in Fig. 1.

Conditional probabilities are definitional. For example,

$$\text{Prob}[G | A] = \frac{g_A}{a} = \frac{g_{LA} + g_{HA}}{a} \text{ and}$$

$$\text{Prob}[L | A] = \frac{\ell_A}{a} = \frac{a - h_A}{a} = 1 - \frac{h_A}{a}$$

The relationship between the values of the probabilities for A and B are related. These relationships and the application of them to juror estimates will be helpful for our analysis. The following three lemmas describe some of these relationships. All three follow from the definitions and the total number of each type.

Lemma 1. $\frac{g_L}{\ell} > \frac{g_{LA}}{\ell_A}$ if and only if $\frac{g_{LB}}{\ell_B} > \frac{g_L}{\ell}$, and likewise $\frac{g_H}{h} > \frac{g_{HA}}{h_A}$ if and only if $\frac{g_{HB}}{h_B} > \frac{g_H}{h}$.

This provides an ordering of the probabilities of guilty. As the population probability $\frac{g_L}{\ell} (\frac{g_H}{h})$ is a convex combination of $\frac{g_{LA}}{\ell_A}$ and $\frac{g_{LB}}{\ell_B}$ ($\frac{g_{HA}}{h_A}$ and $\frac{g_{HB}}{h_B}$), it is between the two.

The next lemma provides a similar result for the probability of H and probability of L, conditional on A or B.

Lemma 2. $\frac{h_A}{a} > \frac{h}{n}$ if and only if $\frac{h_B}{b} > \frac{h}{n}$. Similarly, $\frac{\ell_A}{a} > \frac{\ell}{n}$ if and only if $\frac{\ell_B}{b} > \frac{\ell}{n}$.

¹⁴ This could be another attribute such as socio-economic status. Some of these have a choice component to them. People choose the level of education they invest in. However, there are many factors that influence one's choice of education level and what opportunities are available to them. Many of these factors are correlated with race. There is a sense in which education level, socio-economic status and other attributes can be influenced by self-fulfilling stereotypes.

¹³ Highly educated is used, but socio-economic status is also very compelling.

The third lemma shows that the relationship between the probability of H given A and the probability of H given B are the opposite of the relationship between the probability of L given A and the probability of L given B.

Lemma 3. $\frac{h_A}{a} \leq \frac{h_B}{b}$ if and only if $\frac{\ell_A}{a} \geq \frac{\ell_B}{b}$.

3. Estimated values and analysis

Suppose that a juror does not know the probability that a defendant is guilty given that the defendant is in group A, which is given by $\text{Prob}[G | A]$, and estimates this. Importantly, this probability is intended to capture the propensity for crime. It does not imply that group membership (or race or ethnicity) determines guilt. Assume her estimate uses the true/correct values (for the population as a whole) of $\text{Prob}[G | H]$ and $\text{Prob}[G | L]$ and the true/correct probabilities that a member of group A is in H and in L. The idea is that the juror doesn't know or doesn't use (as might fit with aversive discrimination) the true/correct prior $\text{Prob}[G | A]$ but has a sense of how being in H or L influences the probability that the defendant is guilty. The Appendix contains a discussion of other possible estimates by jurors. The juror's estimate is given by

$$\hat{p}[G | A] = \frac{h_A}{a} \frac{g_H}{h} + \frac{\ell_A}{a} \frac{g_L}{\ell}.$$

Note that the correct value is given by

$$\text{Prob}[G | A] = \frac{g_A}{a} = \frac{h_A}{a} \frac{g_{HA}}{h_A} + \frac{\ell_A}{a} \frac{g_{LA}}{\ell_A} = \frac{1}{a} [g_{HA} + g_{LA}].$$

These differ in that the estimate uses the population values g_H/h and g_L/ℓ instead of the actual values for A g_{HA}/h_A and g_{LA}/ℓ_A , respectively. Unless these happen to be equal, the estimate is incorrect.

The estimated prior probability that type A is guilty is too high (too low) relative to the correct value if and only if the estimated prior probability that type B is guilty is too low (too high) relative to the correct value.

Observation 1. $\hat{p}[G | A] > g_A/a$ if and only if $\hat{p}[G | B] < g_B/b$, and vice versa.

This follows from $h = h_A + h_B$ and $\ell = \ell_A + \ell_B$.

If jurors can be prevented from conditioning on race, the prior used is

$$\text{Prob}[G] = \frac{g}{n} = \frac{h}{n} \frac{g_H}{h} + \frac{\ell}{n} \frac{g_L}{\ell} = \frac{a}{n} \frac{g_A}{a} + \frac{b}{n} \frac{g_B}{b}.$$

The last expression shows that $\frac{g}{n}$ is a convex combination of $\frac{a}{n}$ and $\frac{b}{n}$ so it must be that the value of $\frac{g}{n}$ is between the values of $\frac{a}{n}$ and $\frac{b}{n}$. As discussed above, this is what U.S. law requires and the scope for distorting incentives in primary activity is an important argument for such a policy.¹⁵ However, it is interesting to ask if when focusing only on the case at hand there is an information argument for the policy.

The following lemma will be helpful in addressing this.

Lemma 4. If $\hat{p}[G | A] < \hat{p}[G | B]$, it must be that $\hat{p}[G | A] < \frac{g}{n} < \hat{p}[G | B]$. Similarly, if $\hat{p}[G | B] < \hat{p}[G | A]$, it must be that $\hat{p}[G | B] < \frac{g}{n} < \hat{p}[G | A]$.

That is, $\frac{g}{n}$ is also a convex combination of $\hat{p}[G | A]$ and $\hat{p}[G | B]$.

Observation 2. If $\hat{p}[G | A]$ is too high relative to the correct prior, implying $\hat{p}[G | B]$ is too low, (or vice versa) requiring use of the correct population prior, $\frac{g}{n}$, lowers the prior used for A and raises the prior used for B (or vice versa).

This follows directly from Lemma 4.

In practice this cannot be fully enforced and the extent to which it can be influenced is unclear. However, studies of aversive discrimination suggest that making race salient can have a similar effect. So, it is interesting to consider when attempting to prohibit jurors from conditioning on observable characteristics A or B improves accuracy. The question is essentially whether not conditioning on the defendant's membership in A or B corrects the mistake done by incorrect probability conditioning.

Observation 3. A sufficient condition for $\hat{p}[G | A] > \text{Prob}[G | A]$ is that the majority group is more guilty in both H and L.

That is, the sins of the majority group (group B) are passed by proxy to the minority group (group A).¹⁶ The intuition is the incorrect conditioning is detrimental to the member of group A if A has a lower propensity to crime in both H and L.¹⁷ This follows since the population probabilities $\frac{g_H}{h}$ and $\frac{g_L}{\ell}$ are convex combinations of the corresponding probabilities for group A and group B. When this sufficient condition holds, requiring that jurors not condition on A and B, and instead use the population prior $\frac{g}{n}$ improves accuracy. In fact, this is the case for any situation where $\text{Prob}[G | A] < \text{Prob}[G | B]$ but $\hat{p}[G | A] > \text{Prob}[G | A]$.

Observation 4. Suppose $\text{Prob}[G | A] < \text{Prob}[G | B]$ and $\hat{p}[G | A] > \hat{p}[G | B]$ (or the opposites). Requiring that jurors not condition on A and B, which implies using the population prior, $\frac{g}{n}$, improves accuracy.

By Lemma 4 and that $\frac{g}{n}$ is a convex combination of $\frac{g_A}{a}$ and $\frac{g_B}{b}$, we have $\hat{p}[G | B] < \frac{g}{n} < \hat{p}[G | A]$ and $\text{Prob}[G | A] < \frac{g}{n} < \text{Prob}[G | B]$. Requiring use of the correct population prior $\frac{g}{n}$ lowers the prior used for A and raises the prior used for B. Note that $\frac{g}{n}$ is closer to the correct prior $\text{Prob}[G | A]$ than the estimated prior $\hat{p}[G | A]$, and is closer to the correct prior $\text{Prob}[G | B]$ than the estimated prior $\hat{p}[G | B]$. So this improves the accuracy for both A and B.

Following up on Observation 3, let's consider a necessary and sufficient condition for $\hat{p}[G | A] > \text{Prob}[G | A]$.

Observation 5. A necessary and sufficient condition for $\hat{p}[G | A] > \text{Prob}[G | A]$ is $h_A[\frac{g_H}{h} - \frac{g_{HA}}{h_A}] > -\ell_A[\frac{g_L}{\ell} - \frac{g_{LA}}{\ell_A}]$.

When, as in Observation 3, $\frac{g_H}{h} > \frac{g_{HA}}{h_A}$ and $\frac{g_L}{\ell} > \frac{g_{LA}}{\ell_A}$, we have $h_A[\frac{g_H}{h} - \frac{g_{HA}}{h_A}] > -\ell_A[\frac{g_L}{\ell} - \frac{g_{LA}}{\ell_A}]$, and it must be that $\hat{p}[G | A] > \text{Prob}[G | A]$. In the opposite direction, when $\frac{g_H}{h} < \frac{g_{HA}}{h_A}$ and $\frac{g_L}{\ell} < \frac{g_{LA}}{\ell_A}$, we have $h_A[\frac{g_H}{h} - \frac{g_{HA}}{h_A}] < -\ell_A[\frac{g_L}{\ell} - \frac{g_{LA}}{\ell_A}]$. So, it must be that $\hat{p}[G | A] < \text{Prob}[G | A]$.

When the ordering of these are in opposite directions, the relative magnitudes and the ratio of h_A and ℓ_A determine the ranking of $\hat{p}[G | A]$ and $\text{Prob}[G | A]$.

Some intuition and interpretation can be provided by considering the necessary condition in terms of the conditional probabilities. In these terms, the condition is equivalent to

$$\begin{aligned} & \text{Prob}[H | A][\text{Prob}[G | H] - \text{Prob}[G | H, A]] \\ & > -\text{Prob}[L | A][\text{Prob}[G | L] - \text{Prob}[G | L, A]]. \end{aligned}$$

It's a comparison of the differences between the population probability of guilty and group A's probability of guilty (for H and L) weighted by the probability of H and L for those in group A.

We now turn to the case where the correct prior and estimated prior have the same ordering for A and B. The following is helpful.

¹⁵ Again, this is along the lines of arguments in Sanchirico (2001), and Schrag and Scotchmer (1994).

¹⁶ I thank a reviewer for suggesting this result and interpretation.

¹⁷ I thank another reviewer for suggesting this statement of the intuition.

Observation 6. Suppose $\text{Prob}[G | A] < \text{Prob}[G | B]$ and $\hat{p}[G | A] < \hat{p}[G | B]$ (or the opposites). It must be that either

- (a) $\text{Prob}[G | A] < \hat{p}[G | A] < \frac{g_A}{n} < \hat{p}[G | B] < \text{Prob}[G | B]$
 (b) $\hat{p}[G | A] < \text{Prob}[G | A] < \frac{g_A}{n} < \text{Prob}[G | B] < \hat{p}[G | B]$ (or the opposite orderings).

By [Observation 1](#), $\hat{p}[G | A] > g_A/a$ if and only if $\hat{p}[G | B] < g_B/b$, and vice versa. Both orderings (a) and (b) are possible.

Consider ordering (a) first. Here the population prior, $\frac{g_A}{n}$, is farther from the correct prior for both A and B. So requiring the use of $\frac{g_A}{n}$ reduces accuracy.

Observation 7. Suppose (a) holds so that $\text{Prob}[G | A] < \hat{p}[G | A] < \frac{g_A}{n} < \hat{p}[G | B] < \text{Prob}[G | B]$ (or the opposites). Requiring that jurors not condition on A and B, which implies using the population prior, $\frac{g_A}{n}$, reduces accuracy.

Next consider ordering (b). The implications are less clear cut and depend on the specific values of the probabilities.

Observation 8. Suppose (b) holds so that $\hat{p}[G | A] < \text{Prob}[G | A] < \frac{g_A}{n} < \text{Prob}[G | B] < \hat{p}[G | B]$ (or the opposites). The effects on accuracy of requiring that jurors not condition on A and B, which implies using the population prior, $\frac{g_A}{n}$, depend on the values of the probabilities. It's possible for this to improve or reduce accuracy. It's also possible for it to be improving for the prior used for one group and not for the other.

This depends on whether the correct prior (for both A and B) is closer to the estimated prior or to $\frac{g_A}{n}$.

4. Policy discussion and conclusion

[Observation 4](#) shows that when the estimated priors misestimate enough that the relative magnitudes of the correct priors and estimated priors (for A and B) are reversed, requiring that jurors not condition on A or B improves accuracy. However, when condition (a) in [Observation 6](#) is satisfied, such a requirement reduces accuracy. The effect on accuracy is unclear when condition (b) in [Observation 6](#) is satisfied.

Let's consider how these results connect to policy. The previously mentioned Observations imply that the optimal use of an evidentiary rule depends on both the relative guilt in both groups and juror bias. In [Observation 4](#) juror bias is large enough that the relative magnitudes of the correct priors and estimated priors (for A and B) are reversed. Here, accuracy is improved by prohibition. Naturally, this is similar for [Observation 3](#), which implies [Observation 4](#) holds. An example of more crime in the "majority" group would be native-born residents compared to immigrant residents. By consensus in the literature, immigrants engage in less crime.¹⁸ A common mistaken belief is that immigrants are more likely to engage in crime. If jurors hold this mistaken belief, a prohibition on evidence would improve accuracy.

Now consider intuition for [Observation 8](#). Suppose that those in group B engage in crime more often than those in group A. Suppose, as in condition (b), jurors overestimate the probability of guilt for defendants in group B. Then the effect of the evidentiary rule on accuracy depends on whether the distorted beliefs are in a sense larger in magnitude than the inaccuracy that would be introduced from moving from the true guilt probability for defendants in group B to the overall guilt probability across defendants.

Overall, if jurors are biased against a group with a lower crime rate, then we want to prohibit evidence. If jurors are biased against a group with higher crime rates, then we only want to prohibit evidence if the bias is sufficiently large as in condition (b) of [Observation 6](#). "Sufficiently large" roughly corresponds to the current juror bias being

large enough to outweigh the bias that would be introduced by moving from the true probability of guilt for the group to the pooled probability of guilt across both groups. Juror bias is a key factor for the policy implications.¹⁹

There are very compelling reasons to not permit jurors to condition on observable characteristics, such as race, of the defendant. These include the distortion of incentives in primary activity, issues of fairness, and constitutional requirements.²⁰ However, there is a view that when considering only the case at hand doing so results in discarding predictive information. [Lundberg and Mungan \(2022\)](#), in the case without bias, and [Sullivan \(2019\)](#) are examples of formal analysis suggesting this. There is some sense that such a prohibition limits accuracy to achieve other policy objectives. The results here suggest an accuracy reason for prohibiting jurors from conditioning on observable characteristics of the defendant. This complements the findings of [Lundberg and Mungan](#) for their case of sufficient bias.

Naturally, there are limitations to the improvement of accuracy. [Observation 6](#), condition (a) provides a case where it lowers accuracy. As noted above, it's unclear how effectively jurors can be prohibited from conditioning on the defendant's observable characteristics. There are many practical efforts to reduce the extent to which jurors condition on race. Studies on aversive discrimination suggest that making race salient in the case can reduce decision making that has the effect of treating races differently. Given the many effects of doing so, there is generally a reluctance to allow discussion of race at trial.²¹ So, the results may provide guidance on when to allow attorneys to make race salient. If jurors form priors in a manner that does not consider the allocation of people across different groups in a consistent way, the results of this analysis may not hold. However, such formation of priors may strengthen the need for prohibiting jurors from conditioning on observable characteristics of the defendant.

There are situations where such a prohibition reduces accuracy. Since the correct prior probability the defendant is guilty is not observable, it is difficult to know when each result applies. Experimental research may provide insights for this. Additionally, although the method of juror estimation of priors here seems reasonable, there are others. There is a brief discussion of richer models in the [Appendix](#). Various assumptions about how jurors may form priors are testable.

Relatedly, there is a literature aimed at understanding prior beliefs and stereotypes that focuses primarily on productivity and labor markets. See for example [Bordalo et al. \(2019\)](#), which uses laboratory experiments to study how gender stereotypes influence beliefs about ability of oneself and of others for different categories of knowledge. In general, the issues related to productivity differ from those related to jury assessment of guilt, in part, since guilt and innocence are not observable. Further, many people may not have experience with defendants in a criminal trial and the focus here is on juror bias that may be subconscious as discussed in the Introduction.

However, some elements studied by [Bordalo et al. \(2016\)](#) are related. That study takes a "social cognition approach" to stereotypes and suggests it is closely related to heuristics, as in [Kahneman and Tversky \(1972\)](#), to simplify probability assessments. They focus on Kahneman and Tversky's representativeness heuristic. Their idea is that "The most representative types come to mind first and so are overweighted in judgments". If most representative types were included in a criminal trial scenario, perhaps arrest or conviction rates, as some kind of indication of guilty rates (the correct values of which are not observable), would be the most direct.

¹⁹ I thank a reviewer for encouraging me to more clearly address this and for suggesting these meaningful examples.

²⁰ As noted above, the distortion of incentives in primary activity argument is along the lines of analysis in [Sanchirico \(2001\)](#), and [Schrag and Scotchmer \(1994\)](#).

²¹ See, for example, [Armour \(1995\)](#) for a discussion.

¹⁸ See, for example, [Alesina and Tabellini \(2024\)](#). I thank a reviewer for suggesting this example.

A more realistic approach that fits with aversive discrimination, which has received considerable study in legal psychology, might have to do with differences in educational levels or other socio-economic attributes across groups, which is part of the motivation for this study. As briefly discussed in the [Appendix](#), such erroneous estimates could be included in this analysis with similar results. Here, there is use of a different heuristics in assessing the prior/pre-trial probability of guilt. That is the population probability of guilt given membership in a secondary group (group H or L) is used as an estimate of the group A or B-specific value. Some motivation for this might be that a juror has a better sense of how education level (or some other factor or group membership) influences guilt for the population as a whole and that is first considered and is the value used. One might view this as an adjustment and anchoring heuristic as in [Kahneman and Tversky \(1974\)](#) or as an availability heuristic as in [Kahneman and Tversky \(1973\)](#). I suggest that such an estimate where the correct values are not utilized, which is motivated by aversive discrimination, is reasonable to study, and suggest this is complementary to the work of [Bordalo et al. \(2016\)](#).

I believe better understanding how jurors form priors and make decisions is important. Some of my own ongoing work with collaborators focuses on using laboratory experiments and surveys to do this. An in progress project, which is informed by the exercise reported here, is aimed at testing whether and how subjects use beliefs about a defendant's membership in other groups to form beliefs that the defendant is guilty. Preliminary analysis of some pilot data suggests that subjects may use their beliefs about a minority defendant's education level in evaluating guilt.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Proofs

This appendix contains proofs of the results.

Lemma 1. $\frac{g_L}{\ell} > \frac{g_{LA}}{\ell_A}$ if and only if $\frac{g_{LB}}{\ell_B} > \frac{g_L}{\ell}$, and likewise $\frac{g_H}{h} > \frac{g_{HA}}{h_A}$ if and only if $\frac{g_{HB}}{h_B} > \frac{g_H}{h}$.

Proof. Consider the result for ℓ and L . The construction is analogous for h and H . To show that the second condition implies the first, assume $\frac{g_{LB}}{\ell_B} > \frac{g_L}{\ell}$. This can be rewritten as

$$\frac{g_L - g_{LA}}{\ell - \ell_A} > \frac{g_L}{\ell},$$

which implies

$$g_L \ell - g_{LA} \ell > g_L \ell - g_L \ell_A.$$

or

$$-g_{LA} \ell > -g_L \ell_A.$$

Rearranging yields

$$\frac{g_{LA}}{\ell_A} < \frac{g_L}{\ell}.$$

A similar method shows that the first condition implies the second. Assume $\frac{g_L}{\ell} > \frac{g_{LA}}{\ell_A}$. This can be rewritten as

$$\frac{g_L}{\ell} > \frac{g_L - g_{LB}}{\ell - \ell_B},$$

which implies

$$g_L \ell - g_L \ell_B > g_L \ell - g_{LB} \ell,$$

or

$$-g_{LB} \ell > -g_L \ell_B.$$

Rearranging yields

$$\frac{g_L}{\ell} < \frac{g_{LB}}{\ell_B}. \quad \square$$

Lemma 2. $\frac{h_A}{a} > \frac{h}{n}$ if and only if $\frac{h}{n} > \frac{h_B}{b}$. Similarly, $\frac{\ell_A}{a} > \frac{\ell}{n}$ if and only if $\frac{\ell}{n} > \frac{\ell_B}{b}$.

Proof. Consider the result for ℓ . The construction is analogous for h . To show that the second condition implies the first, assume $\frac{\ell_B}{b} < \frac{\ell}{n}$. This can be rewritten as

$$\frac{\ell - \ell_A}{n - a} < \frac{\ell}{n},$$

which implies

$$n\ell - n\ell_A < n\ell - a\ell,$$

or

$$\frac{\ell_A}{a} > \frac{\ell}{n}.$$

A similar method shows that the first condition implies the second. $\square$

Lemma 3. $\frac{h_A}{a} \leq \frac{h_B}{b}$ if and only if $\frac{\ell_A}{a} \geq \frac{\ell_B}{b}$.

Proof. Note that

$$\frac{\ell_A}{a} = \frac{a - h_A}{a} = 1 - \frac{h_A}{a}$$

and

$$\frac{\ell_B}{b} = \frac{b - h_B}{b} = 1 - \frac{h_B}{b}.$$

So

$$\frac{\ell_A}{a} > \frac{\ell_B}{b} \text{ if and only if } \frac{h_A}{a} < \frac{h_B}{b}.$$

The other direction of inequalities is similar. $\square$

Observation 1. $\hat{p}[G | A] > g_A/a$ if and only if $\hat{p}[G | B] < g_B/b$, and vice versa.

Proof. Suppose not. So we have $\hat{p}[G | A] > g_A/a = \text{Prob}[G | A]$ and $\hat{p}[G | B] \geq g_B/b = \text{Prob}[G | B]$. This implies

$$\frac{h_A}{h} g_H + \frac{\ell_A}{\ell} g_L > g_A \text{ and } [1 - \frac{h_A}{h}] g_H + [1 - \frac{\ell_A}{\ell}] g_L \geq g - g_A.$$

Adding these two expressions yields $g_H + g_L > g$ but it must be that $g_H + g_L = g$. So this is a contradiction. $\square$

Lemma 4. If $\hat{p}[G | A] < \hat{p}[G | B]$, it must be that $\hat{p}[G | A] < \frac{g}{n} < \hat{p}[G | B]$. Similarly, if $\hat{p}[G | B] < \hat{p}[G | A]$, it must be that $\hat{p}[G | B] < \frac{g}{n} < \hat{p}[G | A]$.

Proof. This follows since $\frac{g}{n}$ is a convex combination of $\hat{p}[G | A]$ and $\hat{p}[G | B]$. To show this, note that $\frac{a}{n} + \frac{b}{n} = 1$. We can write

$$\frac{a}{n} \hat{p}[G | A] + \frac{b}{n} \hat{p}[G | B]$$

as

$$\frac{a}{n} \frac{h_A}{a} \frac{g_H}{h} + \frac{a}{n} \frac{\ell_A}{a} \frac{g_L}{\ell} + \frac{b}{n} \frac{h - h_A}{b} \frac{g_H}{h} + \frac{b}{n} \frac{\ell - \ell_A}{b} \frac{g_L}{\ell}$$

or

$$\frac{h_A}{n} \frac{g_H}{h} + \frac{\ell_A}{n} \frac{g_L}{\ell} + \frac{h - h_A}{n} \frac{g_H}{h} + \frac{\ell - \ell_A}{n} \frac{g_L}{\ell}$$

or

$$\frac{h}{n} \frac{g_H}{h} + \frac{\ell}{n} \frac{g_L}{\ell} = \frac{1}{n} [g_H + g_L] = \frac{g}{n}. \quad \square$$

Observation 2. If $\hat{p}[G | A]$ is too high relative to the correct prior, implying $\hat{p}[G | B]$ is too low, (or vice versa) requiring use of the correct population prior $\frac{g}{n}$ lowers the prior used for A and raises the prior used for B (or vice versa).

Proof.

$$\hat{p}[G | A] = \frac{h_A}{a} \frac{g_H}{h} + \frac{\ell_A}{a} \frac{g_L}{\ell},$$

and

$$\hat{p}[G | B] = \frac{h_B}{b} \frac{g_H}{h} + \frac{\ell_B}{b} \frac{g_L}{\ell}.$$

From their definitions, we have $\frac{h_A}{a} > \frac{h}{n}$ if and only if $\frac{h}{n} > \frac{h_B}{b}$. Similarly, $\frac{\ell_A}{a} > \frac{\ell}{n}$ if and only if $\frac{\ell}{n} > \frac{\ell_B}{b}$. Note $\frac{h}{n}$ is a convex combination of $\frac{h_A}{a}$ and $\frac{h_B}{b}$ and its value is between the two. This also applies to $\frac{\ell}{n}$. So if $\hat{p}[G | A]$ is too high relative to the correct prior, requiring the use of $\text{Prob}[G] = \frac{h}{n} \frac{g_H}{h} + \frac{\ell}{n} \frac{g_L}{\ell}$ results in a prior of A being guilty that is lower and a prior of B being guilty that is higher. $\square$

Observation 3. A sufficient condition for $\hat{p}[G | A] > \text{Prob}[G | A]$ is that the majority group is more guilty in both H and L .

Proof. $\frac{g_H}{h} > \frac{g_{HA}}{h_A}$ and $\frac{g_L}{\ell} > \frac{g_{LA}}{\ell_A}$, which by Lemma 1 occur if and only if $\frac{g_{LB}}{\ell_B} > \frac{g_L}{\ell}$ and $\frac{g_{HB}}{h_B} > \frac{g_H}{h}$, imply

$$\frac{h_A}{a} \frac{g_H}{h} + \frac{\ell_A}{a} \frac{g_L}{\ell} > \frac{h_A}{a} \frac{g_{HA}}{h_A} + \frac{\ell_A}{a} \frac{g_{LA}}{\ell_A}.$$

So $\hat{p}[G | A] > \text{Prob}[G | A]$. $\square$

Observation 4. Suppose $\text{Prob}[G | A] < \text{Prob}[G | B]$ and $\hat{p}[G | A] > \hat{p}[G | B]$ (or the opposites). Requiring that jurors not condition on A and B, which implies using the population prior $\frac{g}{n}$ improves accuracy.

Proof. By Lemma 4 and that $\frac{g}{n}$ is a convex combination of $\frac{g_A}{a}$ and $\frac{g_B}{b}$, it must be that

$$\hat{p}[G | B] < \frac{g}{n} < \hat{p}[G | A]$$

and

$$\text{Prob}[G | A] < \frac{g}{n} < \text{Prob}[G | B].$$

So $\frac{g}{n}$ is closer to the value of the correct priors $\text{Prob}[G | A]$ and $\text{Prob}[G | B]$ than the values of the respective estimates $\hat{p}[G | A]$ and $\hat{p}[G | B]$ are. $\square$

Observation 5. A necessary and sufficient condition for $\hat{p}[G | A] > \text{Prob}[G | A]$ is $h_A[\frac{g_H}{h} - \frac{g_{HA}}{h_A}] > -\ell_A[\frac{g_L}{\ell} - \frac{g_{LA}}{\ell_A}]$.

Proof. $\hat{p}[G | A] > \text{Prob}[G | A]$ is equivalent to

$$\frac{h_A}{a} \frac{g_H}{h} + \frac{\ell_A}{a} \frac{g_L}{\ell} > \frac{h_A}{a} \frac{g_{HA}}{h_A} + \frac{\ell_A}{a} \frac{g_{LA}}{\ell_A}.$$

This is equivalent to

$$\frac{h_A}{a} [\frac{g_H}{h} - \frac{g_{HA}}{h_A}] > \frac{\ell_A}{a} [\frac{g_{LA}}{\ell_A} - \frac{g_L}{\ell}].$$

This is equivalent to

$$h_A[\frac{g_H}{h} - \frac{g_{HA}}{h_A}] > -\ell_A[\frac{g_L}{\ell} - \frac{g_{LA}}{\ell_A}]. \quad \square$$

Observation 6. Suppose $\text{Prob}[G | A] < \text{Prob}[G | B]$ and $\hat{p}[G | A] < \hat{p}[G | B]$ (or the opposites). It must be that either

(a) $\text{Prob}[G | A] < \hat{p}[G | A] < \frac{g}{n} < \hat{p}[G | B] < \text{Prob}[G | B]$

(b) $\hat{p}[G | A] < \text{Prob}[G | A] < \frac{g}{n} < \text{Prob}[G | B] < \hat{p}[G | B]$ (or the opposite orderings).

Proof. By Observation 1, $\hat{p}[G | A] > g_A/a$ if and only if $\hat{p}[G | B] < g_B/b$, and vice versa. So, assuming $\text{Prob}[G | A] < \text{Prob}[G | B]$ and $\hat{p}[G | A] < \hat{p}[G | B]$, the following two orderings are possible:

(a) $\text{Prob}[G | A] < \hat{p}[G | A] < \frac{g}{n} < \hat{p}[G | B] < \text{Prob}[G | B]$

and

(b) $\hat{p}[G | A] < \text{Prob}[G | A] < \frac{g}{n} < \text{Prob}[G | B] < \hat{p}[G | B]$.

Examples show that both of these are possible. $\square$

Appendix B. Other approaches

Suppose, instead, that jurors know the correct values of $\frac{g_{HA}}{h_A}$ and $\frac{g_{LA}}{\ell_A}$, but uses the population values for the probability of membership in H and L. So, the estimate of the prior is $\frac{h}{n} \frac{g_{HA}}{h_A} + \frac{\ell}{n} \frac{g_{LA}}{\ell_A}$. In this case Observation 1 does not hold. This is because the population values for membership in H and L are not directly influenced by the probability of guilt given H or L.

A simple example illustrating that Observation 1 does not hold is as follows. Assume $n = 100$, with $a = 40$ and $b = 60$. For A, assume $h_A = 10$, $g_{HA} = 2$, $L_A = 30$ and $g_{LA} = 8$. For B, assume $h_B = 20$, $g_{HB} = 6$, $L_B = 40$, and $g_{LB} = 7$. These values yield $\frac{g_A}{a} = \frac{10}{40}$ and $\frac{g_B}{b} = \frac{13}{60}$. This yields an estimated probability of guilty given A of

$$\bar{p}[G | A] = \frac{30}{100} \frac{2}{10} + \frac{70}{100} \frac{8}{30} = \frac{46}{300} < \frac{1}{4} = \frac{g_A}{a}.$$

Similarly, the estimated probability of guilty given B is

$$\bar{p}[G | B] = \frac{30}{100} \frac{6}{20} + \frac{70}{100} \frac{7}{40} = \frac{79}{400} < \frac{13}{60} = \frac{g_B}{b}.$$

I argue that it's more likely that, when using an estimate of the probability that a member of A or B has a high level of education, jurors use an estimate that underrepresents the fraction of the minority who are highly educated. Jurors may also over estimate the fraction of the majority who are highly educated. Requiring that jurors' estimates of the fraction for A and B who are low education be such that the estimates of low and high education for A (and B) sum to a (and b) provides some consistency that is helpful for proving results.

Adding these sorts of estimates of the education levels of A and B to the model reported in the main text generates similar results to those reported. So for simplicity of exposition, the paper focuses on the simpler setup.

Data availability

No data was used for the research described in the article.

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